THE PICARD GROUP OF A FIRST ORDER THICKENING SITS IN A LONG EXACT SEQUENCE WITH H^I OF THE IDEAL SHEAF

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-more-morphisms/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; the Xy-pic xy environment with its curve arrows is required because the counted statement displays the exact sequence as an Xy-pic diagram. No macro was invented or redefined.

Editorial note. The cross-chapter standard prerequisite is located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim line carrying its canonical tag.

Allowed standard prerequisites (editorial).

- First order thickening with its ideal sheaf. Same chapter, Definition environment labelled definition-thickening (source line 36) and the surrounding discussion of the ideal sheaf of a thickening. Supplies the meaning of the inclusion X contained in X' and of the ideal sheaf I with I^2 = 0; the construction of the exponential sequence is the counted content.
- The Picard group of a ringed space is the first cohomology group of its sheaf of invertible functions. Cohomology, Lemma labelled cohomology-lemma-h1-invertible (canonical Stacks tag 09NU); invoked at proof lines 626-629. Supplies the identification of Pic with H^1 of the sheaf of units, which is what turns the long exact sequence of units into the displayed sequence of Picard groups.

Source reference. Original lemma 09NU: sources/stacks-chow/tags.txt, source lines 12227–12227. Cross-chapter standard prerequisite invoked once at proof lines 626-629 to identify the Picard group with the first cohomology of the sheaf of invertible functions; the exponential sequence and the long exact sequence that produce the displayed diagram are retained verbatim in the counted span.

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Lemma 0.1. *Let $X \subset X'$ be a first order thickening with ideal sheaf $\mathcal{I}$. Then there is a canonical exact sequence*

$$\begin{aligned} 0 \longrightarrow H^0(X, \mathcal{I}) \longrightarrow H^0(X', \mathcal{O}_{X'}^*) \longrightarrow H^0(X, \mathcal{O}_X^*) \\ \longrightarrow H^1(X, \mathcal{I}) \longrightarrow \mathrm{Pic}(X') \longrightarrow \mathrm{Pic}(X) \\ \longrightarrow H^2(X, \mathcal{I}) \longrightarrow \ldots \longrightarrow \ldots \end{aligned}$$

of abelian groups.

Proof. This is the long exact cohomology sequence associated to the short exact sequence of sheaves of abelian groups

$$0 \to \mathcal{I} \to \mathcal{O}_{X'}^* \to \mathcal{O}_X^* \to 0$$

where the first map sends a local section f of $\mathcal{I}$ to the invertible section $1 + f$ of $\mathcal{O}_{X'}$. We also use the identification of the Picard group of a ringed space with the first cohomology group of the sheaf of invertible functions, see Cohomology, Lemma 09NU. □